

Problem B: Sayaka's Blend

Time Limit: 2000ms | Memory Limit: 256MB

Description

"I prefer my coffee precise. Just like my life."

Sayaka Saeki is experimenting with coffee blends at the cafe. She has N distinct types of coffee beans available. The i -th bean has a **Flavor Score** F_i and a **Caffeine Content** C_i .

Sayaka wants to create a blend by selecting a **non-empty** subset of beans. The blend must meet a strict requirement: the **Average Caffeine Content** of the selected beans must be exactly K . Among all valid blends, she wants to maximize the **Total Flavor Score** ($\sum F_i$).

Input

The first line contains two integers N and K ($1 \leq N \leq 40$, $1 \leq K \leq 10^9$). The next N lines each contain two integers F_i and C_i ($1 \leq F_i, C_i \leq 10^9$).

Output

Output the maximum total flavor score possible. If no subset has an average caffeine content of exactly K , output 0.

Examples

Example 1

Input:

```
4 10
100 10
50 20
60 5
40 5
```

Output:

```
250
```

Explanation: We need the average caffeine to be exactly 10. If we select Bean 1 ($C = 10$), Bean 2 ($C = 20$), Bean 3 ($C = 5$), and Bean 4 ($C = 5$): Sum of Caffeine = $10 + 20 + 5 + 5 = 40$. Count = 4. Average = $40/4 = 10$. This is valid. Total Flavor = $100 + 50 + 60 + 40 = 250$.

Example 2

Input:

3 100

10 10

20 20

30 30

Output:

0